

A-5. Fruit Tree Planting

 Agriculture Land Conservation	 High	 High		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions

Measures	Fruit tree planting is an effective agro-ecological landscape restoration measure, contributing to the restoration of degraded lands while enhancing biodiversity, improving soil health, and supporting sustainable livelihoods. It offers a powerful way to restore degraded land, promote biodiversity, and create sustainable livelihoods. By integrating fruit trees into agricultural landscapes, farmers and communities can build resilience to climate change, improve soil health, and ensure long-term food security.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	Suitable for degraded lands, hill slopes								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
			√		√		√		
Feasibility criteria	- Assessing the soil for its fertility, texture, pH, and drainage properties and choosing suitable fruit tree species - Terracing or contour planting to reduce erosion risk, stabilize the soil, and enhance water retention. - Choosing tree species according to the climatic conditions of the area - Considering the investment market return and profitability since forestry products may take years to materialize and result in profit. - Considering market access, value chain and demand of the forestry product as well as transportation and storage facilities.								
Operation and maintenance	Fruit tree plantations in hilly terrain require labor-intensive watering, pruning, pest control, and soil management. Additionally, long establishment period before trees begin producing fruit require significant investment and patience from farmers, with risks from climate variability and potential pest infestations.								
Cost	Medium								
Benefits									
	Type					Low		Medium	High
	Canopy Cover Increase								
	Carbon Sequestration								
	Crop Production Increase								
	Crop Failure Risk Reduction								

	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Fruit trees, especially deep-rooted species, help improve soil structure, reduce erosion, and enhance water retention in the soil. Their roots prevent soil from washing away during heavy rainfall, making them particularly valuable in areas prone to erosion and land degradation. - Planting fruit trees in agro-ecological systems creates diverse habitats for wildlife, including birds, insects, and other beneficial organisms. This			

	promotes biodiversity and can lead to healthier ecosystems that support pollinators and natural pest control agents. • Fruit trees contribute to better water retention and groundwater recharge through their root systems, improving water availability. • Fruit trees enhance nutrient cycling by shedding leaves and organic matter that enrich the soil. Their presence can reduce the need for external fertilizers, promoting more sustainable and organic farming practices.
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.